

Macro Three Replication Project: Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

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Test

1 Replication Target

In this project, we aim to replicate the findings of *From the French franc to the euro, is there an economic impact for the CFA franc zone countries? A 'bias-corrected' synthetic control method* by Alassane Diallo and Adama Ba, published in Cogent Economics & Finance in 2024. This paper assesses the impact of pegging the CFA franc to the euro on real GDP per capita in the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CAEMC) countries. The original study uses a set of 35 annual country-level panel data from 1980 to 2019 and applies synthetic control methods to estimate the counterfactual economic outcomes for member countries.

2 Data

Table 1: Variables				
Variable	Type	Category	(First) Page Ref	Possible Sources
Real GDP per capita	Continuous	Outcome	6	WDI, PennWorld Tables
Lagged GDP per capita	Continuous	Lagged outcome	10	WDI, PennWorld Tables
Gross Fixed Capital Formation (% GDP)	Continuous	Capital	10	
Government expenditure (% GDP)	Continuous	Fiscal	10	WDI, PennWorld Tables
Official Development Assistance (% GDP)	Continuous	External	10	Source not listed
Foreign Development Investment (% GDP)	Continuous	External	10	WDI
Employment rate	Continuous	Labour	10	WDI
Agriculture (% GDP)	Continuous	Structure	10	WDI, PennWorld Tables
Industry (% GDP)	Continuous	Structure	10	WDI, PennWorld Tables
Governance / institutional quality	Index	Institutions	10	Polity5 Dataset
Political regime characteristics	Categorical/Index	Institutions	10	Polity5 Dataset
Inflation	Continuous	Macro controls	5	WDI, PennWorld Tables
Euro peg dummy	Binary	Treatment		Authors
Synthetic GDP per capita	$\tilde{Y}_{it}$	Continuous	Constructed	
Treatment effect	τ_{it}	Continuous	Constructed	

Table 2: CFA franc zone countries annual inflation rate: WAEMU

country	1980–2001	2002–2021
Benin	7.5270	2.1268
Burkina Faso	4.3427	2.2660
Côte d'Ivoire	8.5558	2.1842
Guinea-Bissau	39.2343	2.8921
Mali	8.6642	3.5490
Niger	6.0235	2.8630
Senegal	5.1354	1.8448
Togo	6.2716	2.0069
Average inflation for the WAEMU zone	10.7193	2.4666
Average inflation without Guinea-Bissau	6.6457	2.4058

Table 3: CFA franc zone countries annual inflation rate: CAEMC

country	1980–2001	2002–2021
Cameroon	7.0840	1.8268
Central African Republic	6.5481	2.9430
Chad	5.2621	5.4660
Republic of Congo	7.3655	6.1262
Equatorial Guinea	11.0392	6.7586
Gabon	7.4425	3.7486
Average inflation for the CAEMC zone	7.4569	4.4782

Table 4: Non-CFA franc zone countries mean annual inflation rate

country	1980–2001	2002–2021
Angola	550.9314	32.1383
Burundi	10.3255	9.2785
Congo, Dem. Rep.	1810.0588	16.3427
Ethiopia	5.2079	13.5134
Gambia, The	14.5303	10.1870
Ghana	36.1786	21.0285
Guinea	16.1342	11.2597
Kenya	11.8579	8.5152
Madagascar	18.4232	8.3332
Malawi	23.6956	18.1634
Nigeria	30.4076	13.5749
Sao Tome and Principe	31.6514	10.6113
Sierra Leone	51.3708	11.1235
Sudan	50.6410	37.8117
Tanzania	21.8489	7.4472
Uganda	53.1089	8.7408
Zambia	51.1202	12.4355
Zimbabwe	-1.6381	71.6617
Average inflation	154.7697	17.8981

Table 5: WAEMU countries' mean annual GDP per capita growth: WDI constant and current GDP per capita

country	constant GDP pc growth 1990-2001	constant GDP pc growth 2002-2021	current GDP pc growth 1990-2001	current GDP pc growth 2002-2021
Benin	1.829	1.479	6.649	5.399
Burkina Faso	2.208	2.650	0.655	6.783
Côte d'Ivoire	-1.441	1.457	2.735	5.218
Guinea-Bissau	1.022	1.130	-3.795	5.720
Mali	1.746	0.675	1.871	5.863
Niger	-1.435	1.294	-0.094	5.874
Senegal	0.346	1.435	-1.367	5.024
Togo	-0.422	1.136	-0.773	4.810
Average for the WAEMU area	0.482	1.407	0.735	5.586
Average without Guinea-Bissau	0.404	1.447	1.382	5.567

Table 6: CAEMC countries' mean annual GDP per capita growth: WDI constant and current GDP per capita

country	constant GDP pc growth 1990-2001	constant GDP pc growth 2002-2021	current GDP pc growth 1990-2001	current GDP pc growth 2002-2021
Cameroon	-1.903	1.066	-0.893	4.667
Central African Republic	-1.600	-0.080	-3.905	4.324
Chad	-0.752	2.316	-0.840	10.432
Republic of Congo	-1.257	-0.795	0.293	7.294
Equatorial Guinea	29.871	0.213	26.017	10.281
Gabon	-0.549	-0.728	0.111	4.900
Average for the CAEMC zone	3.968	0.332	3.464	6.983

Table 7: Non-CFA countries' mean annual GDP per capita growth: WDI constant and current GDP per capita

country	constant GDP pc growth 1990-2001	constant GDP pc growth 2002-2021	current GDP pc growth 1990-2001	current GDP pc growth 2002-2021
Angola	-1.741	1.432	-1.873	11.947
Burundi	-2.667	0.049	-3.251	3.790
Congo, Dem. Rep.	-8.290	2.108	13.317	8.093
Ethiopia	-0.075	5.781	-5.088	11.147
Gambia, The	0.221	0.157	7.235	4.525
Ghana	1.661	3.462	-1.294	13.598
Guinea	1.212	2.247	-1.140	5.583
Kenya	-0.764	1.768	2.087	8.684
Madagascar	-0.927	-0.393	2.322	2.806
Malawi	1.172	1.959	2.170	7.383
Nigeria	0.153	2.152	9.192	9.814
Sao Tome and Principe	-0.324	1.953	-1.862	8.234
Sierra Leone	-3.411	2.642	9.782	5.146
Sudan	2.946	-0.546	6.773	3.692
Tanzania	0.851	3.101	4.056	5.696
Uganda	3.159	2.919	-0.793	7.755
Zambia	-0.721	2.231	-1.727	7.007
Zimbabwe	0.660	-0.771	-2.472	13.317
Average	-0.382	1.792	2.080	7.679

Table 8: Estimated synthetic control weights by treated country

Donor country	Benin	Burkina Faso	Côte d'Ivoire	Mali	Niger	Senegal	Togo	Cameroon	Central African Republic	Chad	Republic of Congo	Gabon	Equatorial Guinea
Bangladesh	.771	.901	.353	.946	.757	.356	0	.072	.785	.934	0	0	0
Barbados	.002	0	0	0	0	0	0	0	0	0	0	.345	0
Bhutan	.124	0	0	0	0	0	0	.053	0	0	0	0	.988
Bolivia	0	0	.093	0	0	.261	.011	.002	0	0	.075	0	0
Botswana	0	0	0	0	0	0	0	0	0	0	.007	0	0
Cabo Verde	0	0	0	0	0	0	0	0	0	0	0	0	0
Dominica	0	0	0	0	0	0	0	0	0	0	0	0	.012
Ecuador	.001	0	.110	0	0	0	0	.013	0	0	.159	0	0
Eswatini	0	0	0	.008	0	0	0	0	0	0	0	0	0
Grenada	0	0	0	0	0	0	0	0	0	0	0	0	0
Lao PDR	.094	.097	.120	.040	.236	.047	.301	.017	.191	.066	0	0	0
Lesotho	0	0	0	0	0	.203	.486	0	0	0	0	0	0
Mauritius	0	0	0	0	0	0	0	.083	0	0	0	0	0
Morocco	0	0	.225	0	0	0	0	0	0	0	0	0	0
Namibia	0	0	.074	0	0	.090	0	0	0	0	0	.407	0
St. Kitts and Nevis	0	0	0	0	0	0	0	0	0	0	0	0	0
St. Lucia	0	0	0	0	0	0	0	0	0	0	0	0	0
Seychelles	0	0	0	0	0	0	0	0	0	0	0	0	0
Turkmenistan	0	0	0	0	0	0	.187	.738	0	0	.725	0	0
Guyana	0	0	0	0	0	0	0	0	0	0	0	0	0
Iraq	.008	.002	.026	.006	.008	.043	.015	.022	.023	0	.033	.249	0

Table 9: The average of the predictors of per-capita GDP for franc-CFA zone countries

country	avg_agriculture	avg_government	avg_oda	avg_fdi	avg_labour	avg_institution
Benin	29.961610	0.1633115	54778167	0.4055924	35.62389	2.925000
Synthetic Benin	24.046460	0.1368013	279752921	0.7386912	35.40573	-0.597575
Burkina Faso	27.009303	0.1292807	137699348	0.6938516	40.17606	-2.325000
Synthetic Burkina Faso	24.113678	0.1115558	324146189	0.7302257	34.87846	0.188375
Côte d'Ivoire	21.244944	0.1547087	58697793	1.1245368	31.40414	-2.600000
Synthetic Côte d'Ivoire	18.671979	0.1611628	152098989	1.5637902	35.37862	-0.138075
Mali	31.924485	0.2248145	130880367	1.4313488	30.88820	1.650000
Synthetic Mali	23.398921	0.1071283	338448055	0.5959800	33.96791	0.550050
Niger	36.071288	0.2076225	133696937	2.2883942	41.15699	1.125000
Synthetic Niger	25.420032	0.1262732	279822237	1.1014510	36.74512	-0.958325
Senegal	16.892927	0.1675746	72103592	1.3178785	33.58079	3.150000
Synthetic Senegal	16.282669	0.1718504	152571094	1.8662555	35.34647	2.725000
Togo	21.379832	0.1584141	31225374	1.4919279	40.15292	-3.775000
Synthetic Togo	18.256600	0.2420278	27556161	3.0518995	39.29024	-2.537175
Cameroon	19.749879	0.1512596	53866423	1.2173302	39.96706	-5.600000
Synthetic Cameroon	18.949749	0.2934358	32124099	3.7124711	43.11401	-5.807775
Central African Republic	37.435338	0.1980131	55865711	0.8982344	39.26813	-0.625000
Synthetic Central African Republic	24.722553	0.1231565	290418608	0.9711322	35.87492	-0.699275
Chad	34.548047	0.0753845	72635846	4.1151560	38.32956	-1.850000
Synthetic Chad	23.836711	0.1082252	334233918	0.6479046	34.47783	0.448650
Republic of Congo	7.863290	0.3325782	14367386	5.8508828	33.37802	-4.125000
Synthetic Republic of Congo	17.744385	0.2940493	11537614	3.8061232	43.27624	-4.615925
Gabon	6.291786	0.1734423	1457547	2.4415752	30.06796	-3.375000
Synthetic Gabon	6.789081	0.1871184	43748213	2.7948043	30.81468	1.060050
Equatorial Guinea	6.788717	0.3130109	7265594	17.4621540	33.56924	-6.250000
Synthetic Equatorial Guinea	23.841548	0.3014595	6797661	0.6347582	38.52312	-5.211700

Table 10: The average of the predictors of per-capita GDP for franc-CFA zone countries

country	avg_agriculture	avg_government	avg_oda	avg_fdi	avg_labour	avg_institution
Benin	29.961610	0.1633115	54778167	0.4055924	35.62389	2.925000
Synthetic Benin	24.046460	0.1368013	279752921	0.7386912	35.40573	-0.597575
Burkina Faso	27.009303	0.1292807	137699348	0.6938516	40.17606	-2.325000
Synthetic Burkina Faso	24.113678	0.1115558	324146189	0.7302257	34.87846	0.188375
Côte d'Ivoire	21.244944	0.1547087	58697793	1.1245368	31.40414	-2.600000
Synthetic Côte d'Ivoire	18.671979	0.1611628	152098989	1.5637902	35.37862	-0.138075
Mali	31.924485	0.2248145	130880367	1.4313488	30.88820	1.650000
Synthetic Mali	23.398921	0.1071283	338448055	0.5959800	33.96791	0.550050
Niger	36.071288	0.2076225	133696937	2.2883942	41.15699	1.125000
Synthetic Niger	25.420032	0.1262732	279822237	1.1014510	36.74512	-0.958325
Senegal	16.892927	0.1675746	72103592	1.3178785	33.58079	3.150000
Synthetic Senegal	16.282669	0.1718504	152571094	1.8662555	35.34647	2.725000
Togo	21.379832	0.1584141	31225374	1.4919279	40.15292	-3.775000
Synthetic Togo	18.256600	0.2420278	27556161	3.0518995	39.29024	-2.537175
Cameroon	19.749879	0.1512596	53866423	1.2173302	39.96706	-5.600000
Synthetic Cameroon	18.949749	0.2934358	32124099	3.7124711	43.11401	-5.807775
Central African Republic	37.435338	0.1980131	55865711	0.8982344	39.26813	-0.625000
Synthetic Central African Republic	24.722553	0.1231565	290418608	0.9711322	35.87492	-0.699275
Chad	34.548047	0.0753845	72635846	4.1151560	38.32956	-1.850000
Synthetic Chad	23.836711	0.1082252	334233918	0.6479046	34.47783	0.448650
Republic of Congo	7.863290	0.3325782	14367386	5.8508828	33.37802	-4.125000
Synthetic Republic of Congo	17.744385	0.2940493	11537614	3.8061232	43.27624	-4.615925
Gabon	6.291786	0.1734423	1457547	2.4415752	30.06796	-3.375000
Synthetic Gabon	6.789081	0.1871184	43748213	2.7948043	30.81468	1.060050
Equatorial Guinea	6.788717	0.3130109	7265594	17.4621540	33.56924	-6.250000
Synthetic Equatorial Guinea	23.841548	0.3014595	6797661	0.6347582	38.52312	-5.211700

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(Intercept)	-6346.853 *** (0.000)	922.201 (0.416)	-3750.698 (0.056)	-1066.466 (0.190)	-3922.411 *** (0.000)	-3518.606 * (0.021)	-2892.404 (0.061)	1738.682 (0.198)	-3306.369 *** (0.000)	-7933.350 *** (0.000)	-5861.626 *** (0.000)	-2077.596 (0.178)	-14766.452 (0.305)
treatment	16.353 (0.887)	-151.032 (0.463)	858.879 * (0.038)	457.487 * (0.031)	-288.486 (0.159)	-733.165 * (0.011)	-138.972 (0.508)	-561.236 * (0.011)	-460.778 *** (0.000)	-201.616 (0.262)	-16.122 (0.978)	6052.503 *** (0.000)	19743.264 *** (0.000)
agriculture	-84.797 *** (0.000)	-143.299 *** (0.000)	24.747 (0.316)	-76.185 *** (0.000)	-51.775 *** (0.000)	-183.453 *** (0.000)	-21.035 (0.265)	-170.004 *** (0.000)	-86.113 *** (0.000)	-51.269 *** (0.000)	-125.733 *** (0.000)	-118.040 * (0.030)	-187.621 (0.508)
govt_share	420.284 (0.706)	2647.797 (0.404)	-7217.829 ** (0.005)	1254.588 (0.181)	-2062.144 (0.271)	4997.401 ** (0.003)	-9801.222 *** (0.000)	5943.728 * (0.026)	-4290.706 *** (0.000)	10212.316 *** (0.000)	-112.792 (0.944)	-33025.860 *** (0.000)	-20146.675 * (0.034)
invest_share	-5522.461 ** (0.004)	1186.049 (0.545)	9795.311 *** (0.000)	-6531.285 * (0.013)	6214.008 *** (0.000)	1344.014 (0.563)	5757.113 *** (0.000)	6851.005 ** (0.004)	-4715.750 *** (0.001)	5175.779 *** (0.000)	1226.291 (0.368)	4180.396 (0.235)	-3962.868 (0.573)
oda_share	0.000 *** (0.000)	0.000 *** (0.000)	-0.000 *** (0.000)	-0.000 * (0.013)	0.000 *** (0.000)	0.000 (0.643)	0.000 *** (0.000)	0.000 (0.286)	0.000 *** (0.000)	0.000 *** (0.001)	-0.000 ** (0.004)	-0.000 ** (0.002)	-0.000 (0.769)
fdi	241.086 *** (0.000)	189.812 * (0.021)	-469.429 *** (0.000)	213.082 *** (0.000)	-92.868 ** (0.003)	-217.459 ** (0.007)	-42.890 (0.300)	-132.723 (0.132)	-5.535 (0.902)	-53.509 *** (0.000)	1.703 (0.934)	-90.464 (0.143)	-11.628 (0.638)
labour	320.790 *** (0.000)	82.210 *** (0.000)	209.336 *** (0.000)	176.143 *** (0.000)	155.577 *** (0.000)	267.790 *** (0.000)	140.721 *** (0.000)	76.069 * (0.023)	232.405 *** (0.000)	256.101 *** (0.000)	322.519 *** (0.000)	633.339 *** (0.000)	869.729 * (0.029)
N	80	80	80	80	80	80	80	80	80	80	80	80	80
R2 adj.	0.842	0.799	0.759	0.862	0.774	0.826	0.593	0.694	0.885	0.836	0.744	0.820	0.848

*** p < 0.001; ** p < 0.01; * p < 0.05.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(Intercept)	-11347.371 ** (0.003)	1129.383 *** (0.000)	5342.092 *** (0.000)	-4444.121 *** (0.000)	2437.194 *** (0.000)	-3683.026 *** (0.000)	4393.239 ** (0.004)	-3619.401 ** (0.009)	6772.539 *** (0.000)	7149.720 *** (0.000)	-6705.858 * (0.012)	28646.654 *** (0.000)	-32121.865 * (0.015)
agriculture	-34.223 *** (0.000)	-6.588 (0.186)	1.799 (0.788)	3.341 (0.525)	1.295 (0.596)	18.155 (0.148)	-11.259 * (0.034)	-54.536 *** (0.000)	-8.770 (0.108)	2.194 (0.661)	-45.280 (0.409)	-432.468 ** (0.002)	-125.225 (0.603)
govt_share	-185.268 (0.761)	1472.474 (0.140)	-1183.296 (0.171)	465.725 (0.163)	283.510 (0.337)	2303.524 *** (0.000)	-251.554 (0.832)	-3761.873 * (0.019)	-68.385 (0.860)	6123.254 *** (0.001)	-934.792 (0.362)	-18621.517 *** (0.000)	-9147.003 (0.245)
invest_share	-1350.528 (0.179)	1320.611 ** (0.006)	3103.368 *** (0.000)	6137.186 *** (0.000)	453.508 * (0.048)	3221.948 *** (0.000)	-306.931 (0.429)	1218.417 (0.306)	1427.160 *** (0.000)	1825.971 ** (0.002)	1296.740 (0.112)	9044.053 ** (0.009)	23057.961 * (0.029)
oda_share	0.000 (0.080)	-0.000 (0.866)	-0.000 ** (0.003)	0.000 (0.573)	-0.000 (0.162)	0.000 (0.473)	0.000 (0.147)	0.000 (0.293)	-0.000 *** (0.000)	-0.000 (0.083)	-0.000 (0.946)	0.000 (0.848)	0.000 (0.758)
fdi	179.398 *** (0.000)	29.674 * (0.036)	79.559 (0.162)	31.053 (0.156)	15.976 ** (0.001)	-39.253 (0.099)	21.806 * (0.044)	-17.398 (0.629)	23.397 * (0.025)	-17.951 *** (0.000)	5.469 (0.556)	180.931 *** (0.000)	-63.814 ** (0.001)
labour	363.775 *** (0.001)	-23.535 *** (0.000)	-135.885 *** (0.000)	125.022 *** (0.000)	-54.596 *** (0.000)	104.151 *** (0.000)	-89.867 * (0.011)	153.302 *** (0.000)	-155.978 ** (0.002)	-187.651 *** (0.000)	259.913 ** (0.001)	-628.157 *** (0.000)	1145.132 ** (0.005)
N	40	40	40	40	40	40	40	40	40	40	40	40	40
R2 adj.	0.810	0.924	0.883	0.729	0.856	0.890	0.305	0.541	0.539	0.919	0.698	0.856	0.786

*** p < 0.001; ** p < 0.01; * p < 0.05.

Table 11: Extension SCM summary for trade outcomes

outcome	treated_iso3c	treated_country	pre_rmspe	post_rmspe	rmspe_ratio	avg_post_gap	avg_abs_post_gap	outcome_label
MEU	BEN	Benin	4.740438	15.617041	3.2944296	-13.8408334	13.840833	Imports from EU (% of total imports)
MEU	BFA	Burkina Faso	5.205163	14.847303	2.8524186	-14.3512625	14.351263	Imports from EU (% of total imports)
MEU	CAF	Central African Republic	12.029886	24.439553	2.0315698	-21.1949729	21.714439	Imports from EU (% of total imports)
MEU	CIV	Côte d'Ivoire	5.132861	18.955975	3.6930623	-17.1880613	17.188061	Imports from EU (% of total imports)
MEU	CMR	Cameroon	9.125855	26.179158	2.8686799	-25.0141587	25.014159	Imports from EU (% of total imports)
MEU	COG	Republic of Congo	14.375437	37.836069	2.6319943	-36.2932688	36.293269	Imports from EU (% of total imports)
MEU	GAB	Gabon	7.555036	30.751044	4.0702711	-21.5499129	24.434454	Imports from EU (% of total imports)
MEU	GNQ	Equatorial Guinea	17.363994	23.870283	1.3747001	-23.0139310	23.013931	Imports from EU (% of total imports)
MEU	MLI	Mali	10.233356	10.643753	1.0401039	-9.6770227	9.677023	Imports from EU (% of total imports)
MEU	NER	Niger	7.016746	9.172159	1.3071812	-8.0933944	8.610415	Imports from EU (% of total imports)
MEU	SEN	Senegal	3.668794	12.681645	3.4566250	-11.8079709	11.807971	Imports from EU (% of total imports)
MEU	TCD	Chad	10.329470	25.724796	2.4904275	-23.1336859	23.262100	Imports from EU (% of total imports)
MEU	TGO	Togo	10.970151	16.456914	1.5001538	-14.3598300	14.359830	Imports from EU (% of total imports)
XEU	BEN	Benin	20.203927	16.245284	0.8040657	-16.0536464	16.053646	Exports to EU (% of total exports)
XEU	BFA	Burkina Faso	8.860764	24.476528	2.7623495	-21.8872756	23.559216	Exports to EU (% of total exports)
XEU	CAF	Central African Republic	22.332967	16.361209	0.7326035	-3.7435440	13.077108	Exports to EU (% of total exports)
XEU	CIV	Côte d'Ivoire	6.277816	14.528041	2.3141872	-13.8755376	13.875538	Exports to EU (% of total exports)
XEU	CMR	Cameroon	14.318287	6.591509	0.4603560	-1.0068068	5.260558	Exports to EU (% of total exports)
XEU	COG	Republic of Congo	13.225655	13.642517	1.0315191	-10.4144214	12.755337	Exports to EU (% of total exports)
XEU	GAB	Gabon	15.327433	12.923145	0.8431383	-11.8499627	11.849963	Exports to EU (% of total exports)
XEU	GNQ	Equatorial Guinea	29.864246	32.023694	1.0723088	-30.8151028	30.815103	Exports to EU (% of total exports)
XEU	MLI	Mali	14.845509	26.343900	1.7745367	-25.8672286	25.867229	Exports to EU (% of total exports)
XEU	NER	Niger	12.137954	20.813455	1.7147416	-18.4445869	18.444587	Exports to EU (% of total exports)
XEU	SEN	Senegal	5.393509	24.955852	4.6270159	-24.2435584	24.243558	Exports to EU (% of total exports)
XEU	TCD	Chad	16.434477	49.902716	3.0364652	-47.2601214	47.460121	Exports to EU (% of total exports)
XEU	TGO	Togo	17.258511	16.784060	0.9725092	-16.1284693	16.128469	Exports to EU (% of total exports)
exports_gdp	BEN	Benin	3.361911	6.417701	1.9089446	-6.0602666	6.060267	Exports of goods and services (% of GDP)
exports_gdp	BFA	Burkina Faso	2.598701	6.207541	2.3887094	-5.9584216	5.958422	Exports of goods and services (% of GDP)
exports_gdp	CAF	Central African Republic	2.358274	13.509221	5.7284368	-12.4497160	12.449716	Exports of goods and services (% of GDP)
exports_gdp	CIV	Côte d'Ivoire	5.079209	4.147575	0.8165790	-4.0559981	4.055998	Exports of goods and services (% of GDP)
exports_gdp	CMR	Cameroon	4.228425	9.232743	2.1834943	-8.4956705	8.495670	Exports of goods and services (% of GDP)
exports_gdp	COG	Republic of Congo	10.891082	12.119244	1.1127677	12.0211839	12.021184	Exports of goods and services (% of GDP)
exports_gdp	GAB	Gabon	9.949549	5.526300	0.5554322	-0.1738516	4.987402	Exports of goods and services (% of GDP)
exports_gdp	MLI	Mali	1.644789	2.337057	1.4208854	0.5028868	1.973068	Exports of goods and services (% of GDP)
exports_gdp	NER	Niger	4.106919	10.403599	2.5331885	-9.9852965	9.985297	Exports of goods and services (% of GDP)
exports_gdp	SEN	Senegal	2.031861	6.559557	3.2283497	-5.6486291	5.648629	Exports of goods and services (% of GDP)
exports_gdp	TCD	Chad	2.778024	20.301452	7.3078740	14.0960344	16.925594	Exports of goods and services (% of GDP)
exports_gdp	TGO	Togo	9.188466	12.383933	1.3477693	-12.3162065	12.316206	Exports of goods and services (% of GDP)
imports_gdp	BEN	Benin	3.237705	12.189825	3.7649590	-11.8696181	11.869618	Imports of goods and services (% of GDP)
imports_gdp	BFA	Burkina Faso	2.789236	5.928738	2.1255779	-5.7271288	5.727129	Imports of goods and services (% of GDP)
imports_gdp	CAF	Central African Republic	2.093336	7.181752	3.4307684	-7.1558313	7.155831	Imports of goods and services (% of GDP)
imports_gdp	CIV	Côte d'Ivoire	2.664478	3.360270	1.2611366	-2.9158345	2.915834	Imports of goods and services (% of GDP)
imports_gdp	CMR	Cameroon	4.239115	5.740685	1.3542177	-4.8180605	5.061897	Imports of goods and services (% of GDP)
imports_gdp	COG	Republic of Congo	14.629023	12.141741	0.8299762	-0.8212777	10.073267	Imports of goods and services (% of GDP)
imports_gdp	GAB	Gabon	6.504453	15.281396	2.3493743	-13.8042022	13.804202	Imports of goods and services (% of GDP)
imports_gdp	MLI	Mali	4.218661	6.111241	1.4486211	-5.5183312	5.518331	Imports of goods and services (% of GDP)
imports_gdp	NER	Niger	4.257775	2.201008	0.5169385	-1.5541337	1.554134	Imports of goods and services (% of GDP)
imports_gdp	SEN	Senegal	3.701995	4.113182	1.1110717	4.1069223	4.106922	Imports of goods and services (% of GDP)
imports_gdp	TCD	Chad	5.682360	43.505743	7.6562802	31.1153966	32.568526	Imports of goods and services (% of GDP)
imports_gdp	TGO	Togo	8.467835	3.004984	0.3548705	-1.7871499	2.603954	Imports of goods and services (% of GDP)
trade_openness	BEN	Benin	5.648168	19.699045	3.4876876	-18.8839147	18.883915	Trade openness (% of GDP)
trade_openness	BFA	Burkina Faso	4.301250	11.058926	2.5710960	-10.6249525	10.624952	Trade openness (% of GDP)
trade_openness	CAF	Central African Republic	4.843208	21.335653	4.4052728	-20.5333222	20.533322	Trade openness (% of GDP)
trade_openness	CIV	Côte d'Ivoire	6.974370	7.702623	1.1044185	-7.1014379	7.101438	Trade openness (% of GDP)
trade_openness	CMR	Cameroon	8.472586	11.241547	1.3268142	-10.5937178	10.593718	Trade openness (% of GDP)
trade_openness	COG	Republic of Congo	22.320958	14.290754	0.6402393	9.3890419	11.411153	Trade openness (% of GDP)

Table 12: Extension SCM summary for trade outcomes

outcome	treated_iso3c	treated_country	pre_rmspe	post_rmspe	rmspe_ratio	avg_post_gap	avg_abs_post_gap	outcome_label
MEU	BEN	Benin	4.740438	15.617041	3.2944296	-13.8408334	13.840833	Imports from EU (% of total imports)
MEU	BFA	Burkina Faso	5.205163	14.847303	2.8524186	-14.3512625	14.351263	Imports from EU (% of total imports)
MEU	CAF	Central African Republic	12.029886	24.439553	2.0315698	-21.1949729	21.714439	Imports from EU (% of total imports)
MEU	CIV	Côte d'Ivoire	5.132861	18.955975	3.6930623	-17.1880613	17.188061	Imports from EU (% of total imports)
MEU	CMR	Cameroon	9.125855	26.179158	2.8686799	-25.0141587	25.014159	Imports from EU (% of total imports)
MEU	COG	Republic of Congo	14.375437	37.836069	2.6319943	-36.2932688	36.293269	Imports from EU (% of total imports)
MEU	GAB	Gabon	7.555036	30.751044	4.0702711	-21.5499129	24.434454	Imports from EU (% of total imports)
MEU	GNQ	Equatorial Guinea	17.363994	23.870283	1.3747001	-23.0139310	23.013931	Imports from EU (% of total imports)
MEU	MLI	Mali	10.233356	10.643753	1.0401039	-9.6770227	9.677023	Imports from EU (% of total imports)
MEU	NER	Niger	7.016746	9.172159	1.3071812	-8.0933944	8.610415	Imports from EU (% of total imports)
MEU	SEN	Senegal	3.668794	12.681645	3.4566250	-11.8079709	11.807971	Imports from EU (% of total imports)
MEU	TCD	Chad	10.329470	25.724796	2.4904275	-23.1336859	23.262100	Imports from EU (% of total imports)
MEU	TGO	Togo	10.970151	16.456914	1.5001538	-14.3598300	14.359830	Imports from EU (% of total imports)
XEU	BEN	Benin	20.203927	16.245284	0.8040657	-16.0536464	16.053646	Exports to EU (% of total exports)
XEU	BFA	Burkina Faso	8.860764	24.476528	2.7623495	-21.8872756	23.559216	Exports to EU (% of total exports)
XEU	CAF	Central African Republic	22.332967	16.361209	0.7326035	-3.7435440	13.077108	Exports to EU (% of total exports)
XEU	CIV	Côte d'Ivoire	6.277816	14.528041	2.3141872	-13.8755376	13.875538	Exports to EU (% of total exports)
XEU	CMR	Cameroon	14.318287	6.591509	0.4603560	-1.0068068	5.260558	Exports to EU (% of total exports)
XEU	COG	Republic of Congo	13.225655	13.642517	1.0315191	-10.4144214	12.755337	Exports to EU (% of total exports)
XEU	GAB	Gabon	15.327433	12.923145	0.8431383	-11.8499627	11.849963	Exports to EU (% of total exports)
XEU	GNQ	Equatorial Guinea	29.864246	32.023694	1.0723088	-30.8151028	30.815103	Exports to EU (% of total exports)
XEU	MLI	Mali	14.845509	26.343900	1.7745367	-25.8672286	25.867229	Exports to EU (% of total exports)
XEU	NER	Niger	12.137954	20.813455	1.7147416	-18.4445869	18.444587	Exports to EU (% of total exports)
XEU	SEN	Senegal	5.393509	24.955852	4.6270159	-24.2435584	24.243558	Exports to EU (% of total exports)
XEU	TCD	Chad	16.434477	49.902716	3.0364652	-47.2601214	47.460121	Exports to EU (% of total exports)
XEU	TGO	Togo	17.258511	16.784060	0.9725092	-16.1284693	16.128469	Exports to EU (% of total exports)
exports_gdp	BEN	Benin	3.361911	6.417701	1.9089446	-6.0602666	6.060267	Exports of goods and services (% of GDP)
exports_gdp	BFA	Burkina Faso	2.598701	6.207541	2.3887094	-5.9584216	5.958422	Exports of goods and services (% of GDP)
exports_gdp	CAF	Central African Republic	2.358274	13.509221	5.7284368	-12.4497160	12.449716	Exports of goods and services (% of GDP)
exports_gdp	CIV	Côte d'Ivoire	5.079209	4.147575	0.8165790	-4.0559981	4.055998	Exports of goods and services (% of GDP)
exports_gdp	CMR	Cameroon	4.228425	9.232743	2.1834943	-8.4956705	8.495670	Exports of goods and services (% of GDP)
exports_gdp	COG	Republic of Congo	10.891082	12.119244	1.1127677	12.0211839	12.021184	Exports of goods and services (% of GDP)
exports_gdp	GAB	Gabon	9.949549	5.526300	0.5554322	-0.1738516	4.987402	Exports of goods and services (% of GDP)
exports_gdp	MLI	Mali	1.644789	2.337057	1.4208854	0.5028868	1.973068	Exports of goods and services (% of GDP)
exports_gdp	NER	Niger	4.106919	10.403599	2.5331885	-9.9852965	9.985297	Exports of goods and services (% of GDP)
exports_gdp	SEN	Senegal	2.031861	6.559557	3.2283497	-5.6486291	5.648629	Exports of goods and services (% of GDP)
exports_gdp	TCD	Chad	2.778024	20.301452	7.3078740	14.0960344	16.925594	Exports of goods and services (% of GDP)
exports_gdp	TGO	Togo	9.188466	12.383933	1.3477693	-12.3162065	12.316206	Exports of goods and services (% of GDP)
imports_gdp	BEN	Benin	3.237705	12.189825	3.7649590	-11.8696181	11.869618	Imports of goods and services (% of GDP)
imports_gdp	BFA	Burkina Faso	2.789236	5.928738	2.1255779	-5.7271288	5.727129	Imports of goods and services (% of GDP)
imports_gdp	CAF	Central African Republic	2.093336	7.181752	3.4307684	-7.1558313	7.155831	Imports of goods and services (% of GDP)
imports_gdp	CIV	Côte d'Ivoire	2.664478	3.360270	1.2611366	-2.9158345	2.915834	Imports of goods and services (% of GDP)
imports_gdp	CMR	Cameroon	4.239115	5.740685	1.3542177	-4.8180605	5.061897	Imports of goods and services (% of GDP)
imports_gdp	COG	Republic of Congo	14.629023	12.141741	0.8299762	-0.8212777	10.073267	Imports of goods and services (% of GDP)
imports_gdp	GAB	Gabon	6.504453	15.281396	2.3493743	-13.8042022	13.804202	Imports of goods and services (% of GDP)
imports_gdp	MLI	Mali	4.218661	6.111241	1.4486211	-5.5183312	5.518331	Imports of goods and services (% of GDP)
imports_gdp	NER	Niger	4.257775	2.201008	0.5169385	-1.5541337	1.554134	Imports of goods and services (% of GDP)
imports_gdp	SEN	Senegal	3.701995	4.113182	1.1110717	4.1069223	4.106922	Imports of goods and services (% of GDP)
imports_gdp	TCD	Chad	5.682360	43.505743	7.6562802	31.1153966	32.568526	Imports of goods and services (% of GDP)
imports_gdp	TGO	Togo	8.467835	3.004984	0.3548705	-1.7871499	2.603954	Imports of goods and services (% of GDP)
trade_openness	BEN	Benin	5.648168	19.699045	3.4876876	-18.8839147	18.883915	Trade openness (% of GDP)
trade_openness	BFA	Burkina Faso	4.301250	11.058926	2.5710960	-10.6249525	10.624952	Trade openness (% of GDP)
trade_openness	CAF	Central African Republic	4.843208	21.335653	4.4052728	-20.5333222	20.533322	Trade openness (% of GDP)
trade_openness	CIV	Côte d'Ivoire	6.974370	7.702623	1.1044185	-7.1014379	7.101438	Trade openness (% of GDP)
trade_openness	CMR	Cameroon	8.472586	11.241547	1.3268142	-10.5937178	10.593718	Trade openness (% of GDP)
trade_openness	COG	Republic of Congo	22.320958	14.290754	0.6402393	9.3890419	11.411153	Trade openness (% of GDP)